Explainable AI for Mental Health Classification

- Addressing Bias and Transparency in AI-Based Mental Health Systems

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Introduction & Motivation

- Building on CW1 research into the ethical implications of mental health AI
- Key concerns identified:
  - Safety and reliability for patients
  - Algorithmic bias and fairness
  - Lack of transparency in AI decision-making
  - Data privacy and consent
- The project addresses these concerns using Explainable AI (XAI) techniques

Research Questions

1. Can XAI techniques like SHAP and LIME make mental health AI more transparent?
2. How can we detect and measure bias in the classification of mental health?
3. What mitigation strategies effectively reduce algorithmic bias?
4. How do we balance model performance with fairness requirements?
5. What are the ethical frameworks for the responsible deployment of AI?

Literature Review: XAI Techniques

- SHAP: SHapley Additive exPlanations
  - Game theory-based feature attribution
  - Global and local interpretability provided
  - Consistent and theoretically sound
- LIME: Local Interpretable Model-agnostic Explanations
- Local approximation of black-box models
  - Model-agnostic approach
  - Human-understandable explanations
- Both critical techniques for clinical AI systems (Lundberg et al., 2020)

Literature Review: Bias in Healthcare AI

- Obermeyer et al. (2019): Racial bias in healthcare algorithms
  - Algorithms showed significant disparities across demographic groups
- Rajkomar et al. (2018): Challenges in clinical ML
- Bias amplification in training data
  - Need for fairness-aware machine learning
- Mehrabi et al. (2021): Survey of fairness in ML
  - Various definitions of fairness: demographic parity, equalized odds
  - Trade-offs between accuracy and fairness

Methodology Overview

- Dataset: Mental Health in Tech Survey, 1000 samples
  - Demographics, work environment, mental health history
  - Target: Treatment-seeking behavior
- Approach:

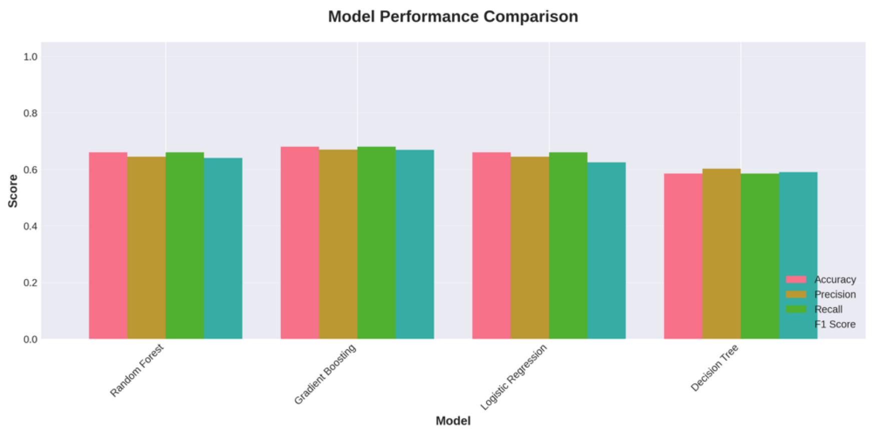
  1. Data preprocessing and exploratory analysis
  2. Multi-model training and evaluation
  3. SHAP and LIME explainability analysis
  4. Bias detection across demographic groups
  5. Bias mitigation through reweighting 6. Ethical implications assessment
  6. Ethical implications assessment

Data Exploration

Distribution of treatment-seeking behavior and age demographics

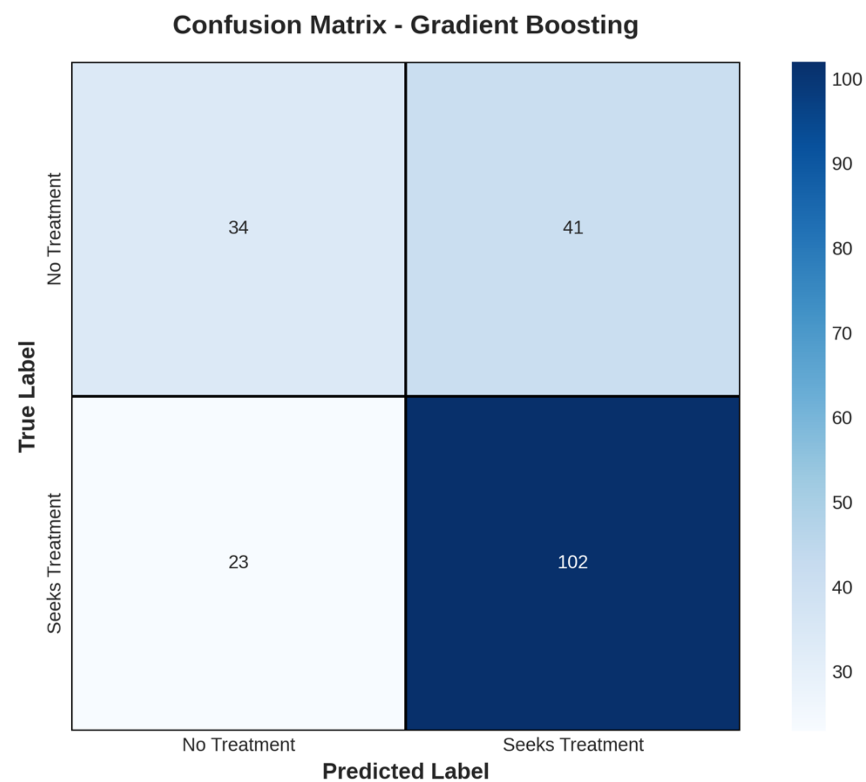
Gradient Boosting achieved best F1 score (0.67)

Model Performance Comparison



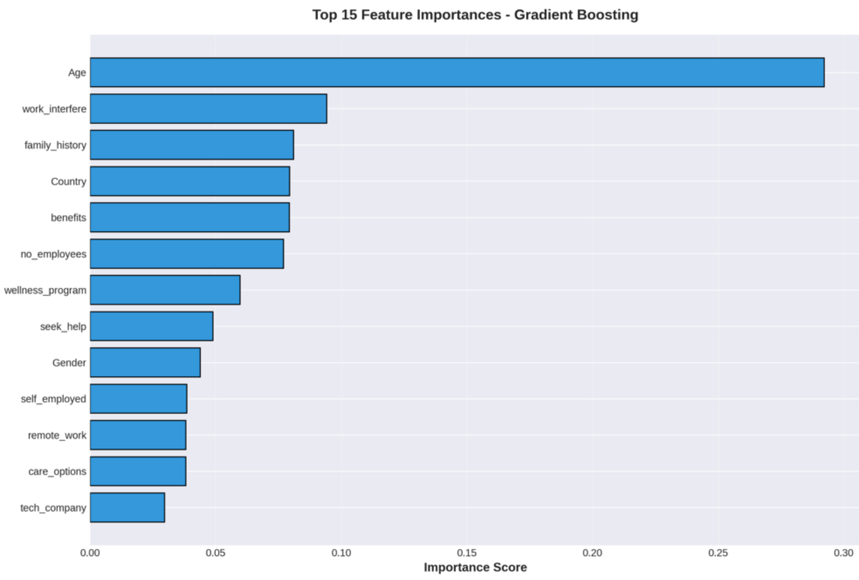
Performance analysis of best performing model

Model Evaluation - Confusion Matrix



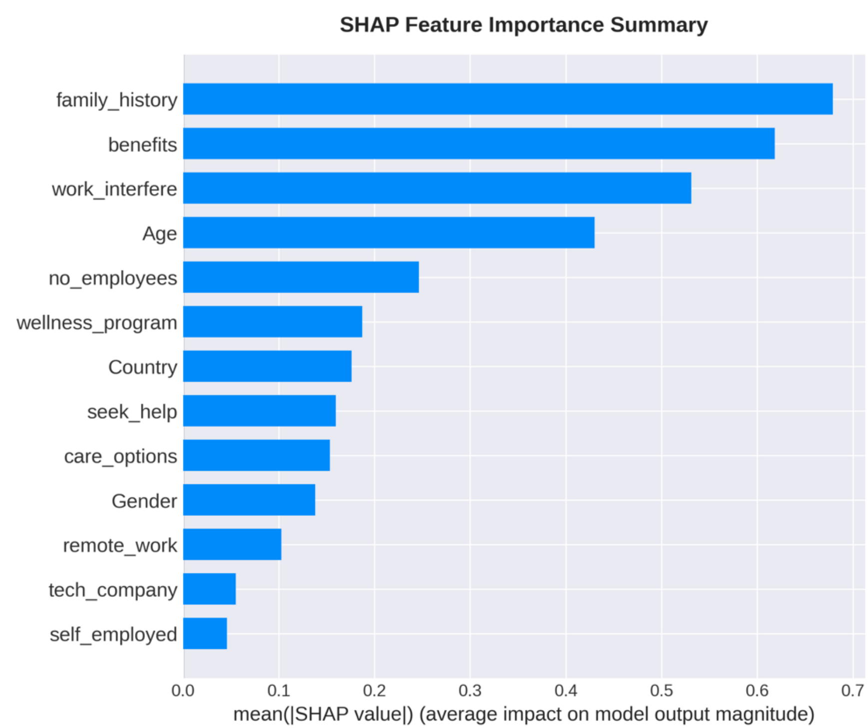
Key predictors: family history, work interference, benefits

Feature Importance Analysis



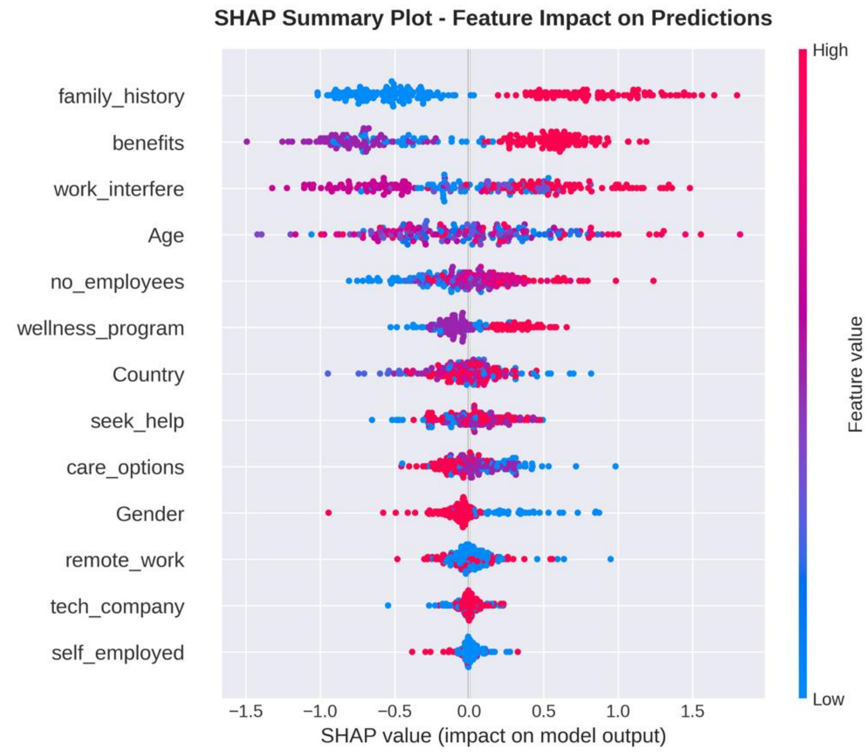
Global feature importance using SHAP values

SHAP Feature Importance



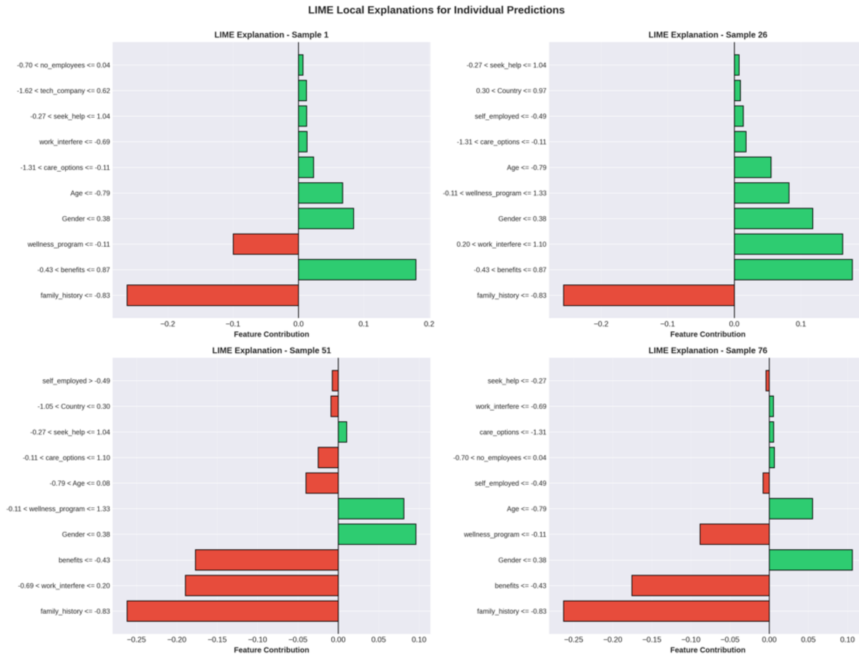
Feature impact on individual predictions

SHAP Detailed Analysis



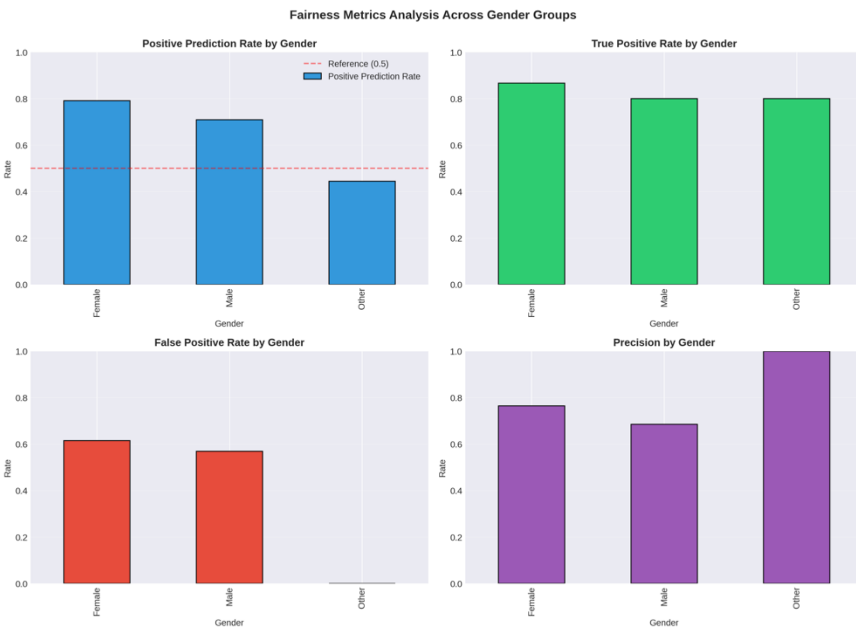
Human-interpretable explanations for individual predictions

LIME Local Explanations



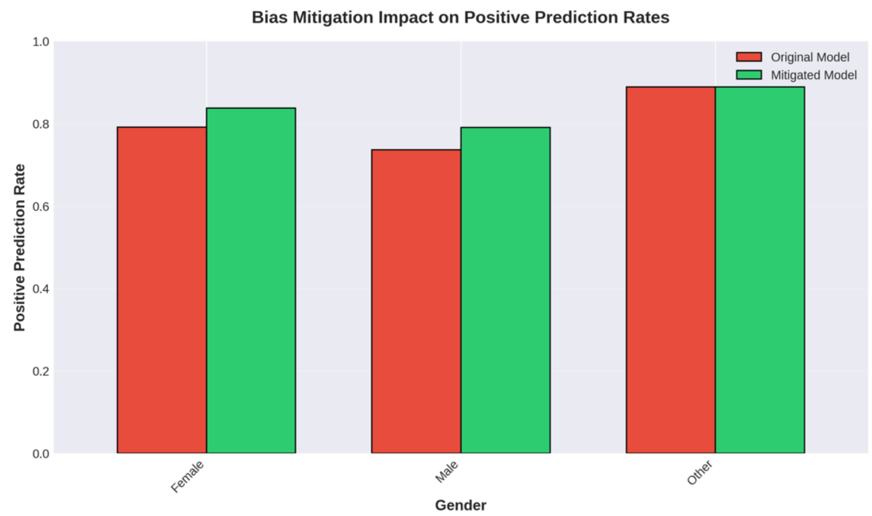
Measuring bias using demographic parity and equalized odds

Fairness Analysis Across Demographics



Impact of reweighting on fairness metrics

Bias Mitigation Results



Key Findings: Explainability

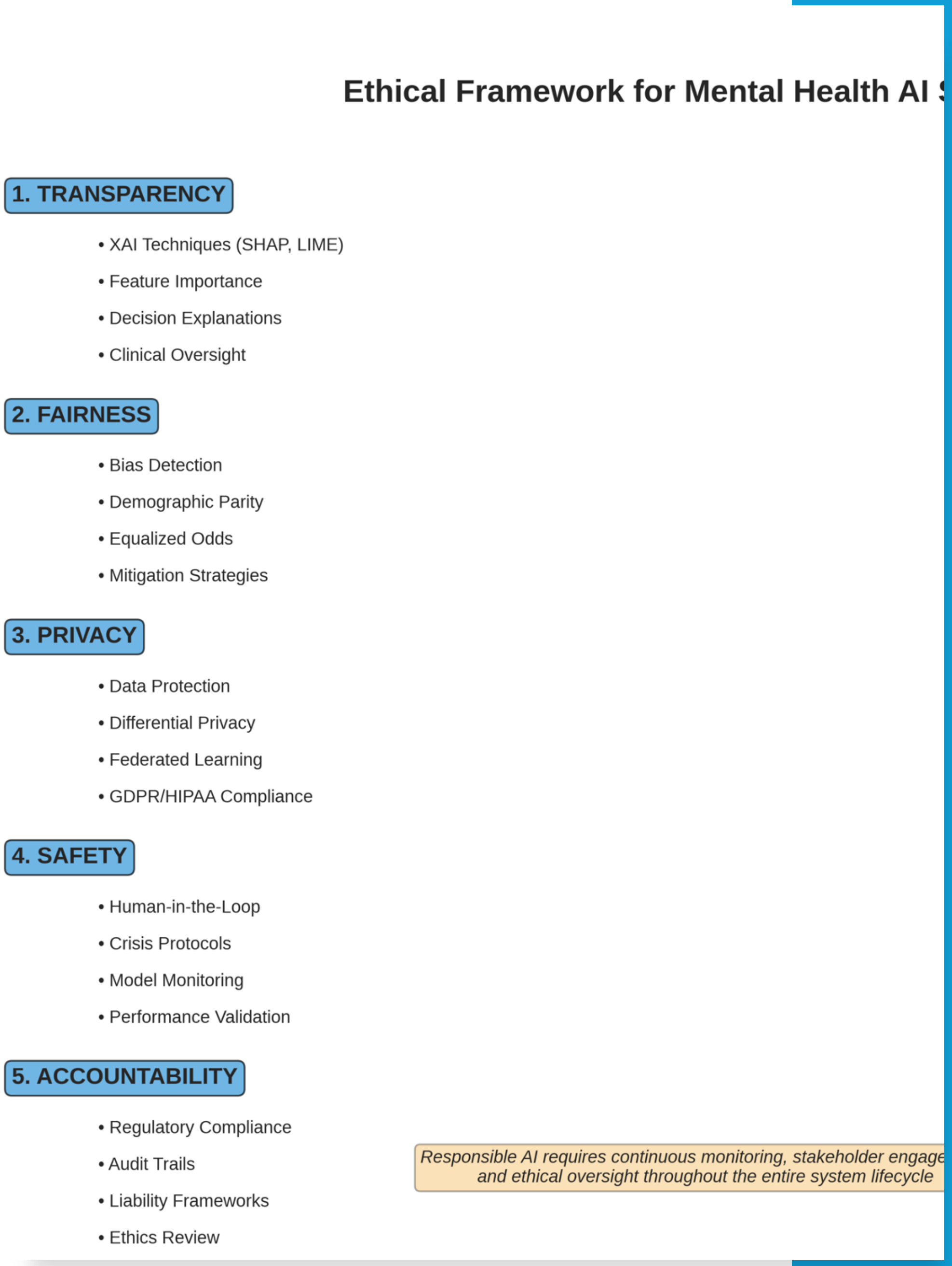
- SHAP Analysis revealed:
  - Family history, work interference, and benefits as top predictors
  - Feature importance is consistent across model types
  - Model explanations for individual predictions are clinically intuitive
- LIME Analysis showed:
  - Local explanations vary significantly between different individuals
- Model-agnostic approach confirms the SHAP results
  - Human-interpretable feature contributions
- Clinical Implications:
  - Enables clinician oversight and validation
  - Informs decision-making

Key Findings: Bias & Fairness

- Bias Detection Results:
  - Disparate Impact Ratio: 1.115 (within acceptable range 0.8-1.25)
  - Statistical parity difference: 0.346 - significant disparity detected.
  - Variable True Positive Rates across Genders
- Mitigation Strategies:
  - Reweighting reduced prediction rate disparities
  - Trade-off: slight loss in overall accuracy
  - Improved fairness metrics across demographics
- Lessons Learned:
  - No single metric of fairness tells the complete story
  - The importance of context-specific fairness definitions

Comprehensive ethical considerations

Ethical Framework for Mental Health AI



Ethical Implications & Recommendations

- Transparency & Explainability:
  - XAI techniques allow for clinical oversight.
  - Individual prediction explanations support informed consent
- Fairness & Non-discrimination:
  - Continuous monitoring across demographic groups was required.
  - Bias mitigation should be context appropriate
- Privacy & Data Protection:
  - GDPR/HIPAA compliance is a must.
  - Differential privacy and federated learning recommended
- Safety & Clinical Oversight:
  - Human-in-the-loop for high-risk decisions
  - Clearly defined escalation procedures for critical situations

Regulatory & Legal Considerations

- Medical Device Regulation:
  - FDA (US) and MHRA (UK) approval pathways
  - Classification as a medical device vs. wellness tool
- AI-Specific Regulation:
  - EU AI Act: Requirements for high-risk AI systems
  - Transparency and accountability requirements
- Liability Frameworks:
  - Unclear assignment of responsibility
  - Requirement for clear contractual agreements
- Data Protection:
  - Cross-border data transfer challenges
  - Patient consent and data rights

Limitations & Future Work

- Current Limitations:
  - Synthetic dataset (demonstration requires real data)
  - Limited demographic diversity
  - Single point-in-time analysis-only, no longitudinal data
  - Cultural bias in feature selection
- Future Work:
  - Expand to multi-modal data: text, voice, behavioral
  - Longitudinal studies of validation
  - Integration with clinical workflows
  - Continuous learning with clinician feedback
  - Cross-cultural validation

Technical Implementation Highlights

- Libraries & Technologies:
  - scikit-learn: Model training and evaluation
  - SHAP: Explainability analysis
  - LIME: Local interpretability
  - pandas, numpy: Data processing
  - matplotlib, seaborn: Visualization
- Code Quality:
  - Modular, well-documented functions
  - Comprehensive error handling
  - Reproducible results (fixed random seeds)
  - Production-ready structure

Total: ~1200 lines of documented Python code

Conclusions

- Demonstrated successful integration of XAI in mental health AI:
  - ✓ SHAP and LIME provide complementary explainability
  - ✓ Bias detection reveals fairness challenges
  - ✓ Mitigation strategies improve equity (with trade-offs)
  - ✓ Comprehensive ethical framework guides deployment
- Key Takeaways:
  - Transparency is achievable through XAI techniques
  - Fairness requires continuous monitoring and context-specific approaches
  - Technical solutions must be coupled with ethical governance
  - Multidisciplinary collaboration is essential
- Mental health AI holds promise, but responsible deployment requires
- ongoing commitment to transparency, fairness, and patient safety

References (1/2)

- Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. NIPS.
- Ribeiro, M. T., et al. (2016). Why should I trust you? Explaining predictions of ML models. KDD.
- Obermeyer, Z., et al. (2019). Dissecting racial bias in an algorithm. Science, 366(6464).
- Rajkomar, A., et al. (2018). Ensuring fairness in ML to advance health equity. Annals of Internal Medicine.
- Mehrabi, N., et al. (2021). A survey on bias and fairness in machine learning. ACM Computing Surveys.

References (2/2)

- Char, D. S., et al. (2020). Implementing ML in health care. NEJM, 383(10).
- Hardt, M., et al. (2016). Equality of opportunity in supervised learning. NIPS.
- Wachter, S., et al. (2021). Bias preservation in ML. West Virginia Law Review.
- Chen, I. Y., et al. (2019). Can AI help reduce disparities in general medical and mental health care? AMA Journal of Ethics.
- Additional references are available in code documentation and ethical report.

Thank You

Questions & Discussion

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